

Pioneering into the Extragalactic Frontier with the GMT
Texas A&M University, 14.03.11

Reionisation and the evolution of the UV background



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the Big Bang

Years after the Big Bang

$z=1100$ 0.3 million

The Universe was filled with ionized gas

Hydrogen turned to be neutral

The Cosmic Dark Age

$z=10$ 0.5 billion

First astronomical objects: formation of galaxies and quasars. Beginning of the cosmic reionization

Epoch of HI reionisation

Renaissance of the Universe - the End of the Dark Age

$z=6$ 1 billion

Completion of the reionization: inter-galactic medium was ionized

Ionized inter-galactic space

Post-reionisation.
UVB keeps the IGM highly ionised.

$z=3$ 2 billion

(The epoch observed in the present research)

Evolution of galaxies

+He II reionisation

$z=1.5$ 9 billion

Formation of the solar system

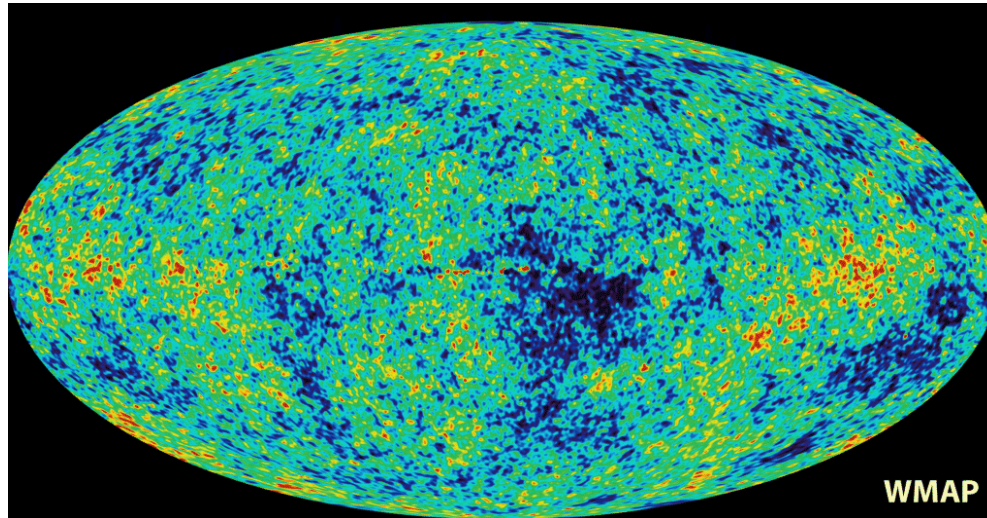
$z=0$ 13.5 billion

The Present Universe

images credit: NASA / WMAP Science Team, Subaru Telescope/NAOJ

The key evidence

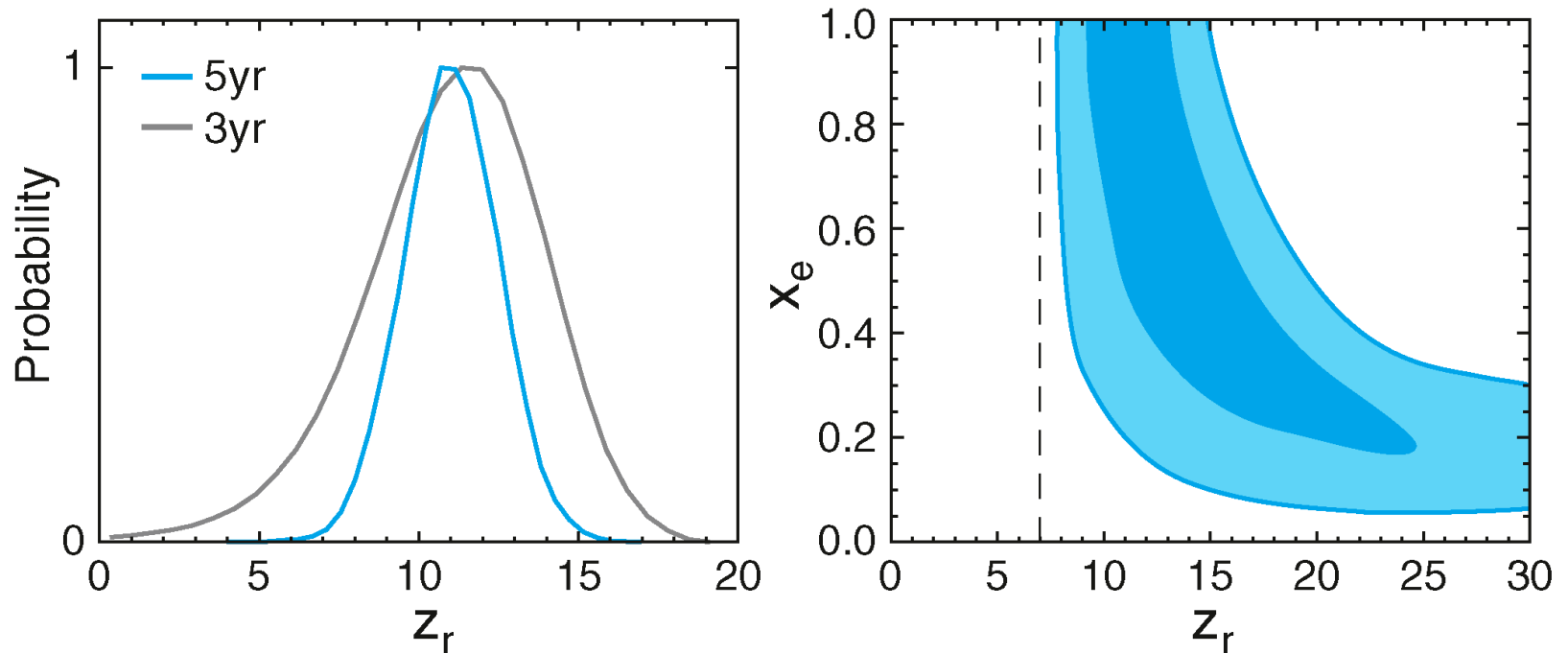
I: Cosmic Microwave Background data



Thomson scattering of CMB photons off free electrons modifies the temperature and temperature-polarisation power spectra

CMB constraint

This only gives a constraint on the integrated reionisation history

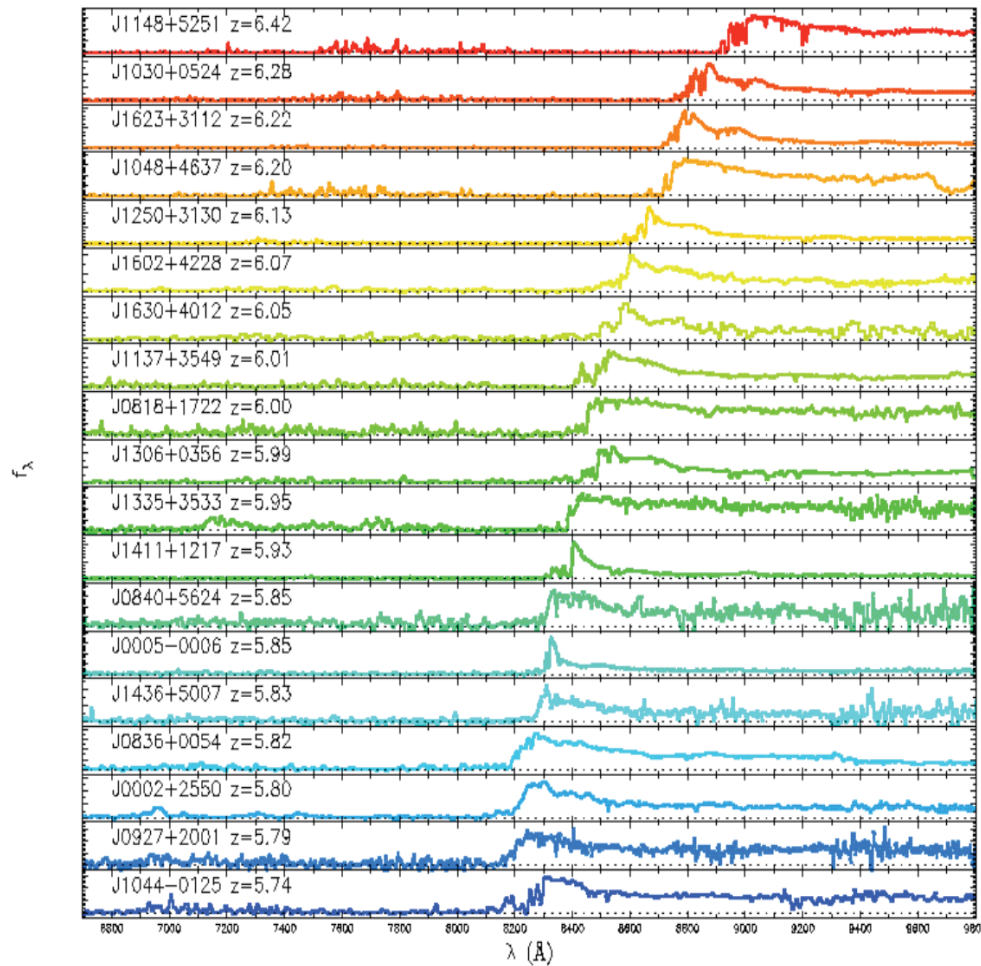


In the limit of instantaneous reionisation (7yr):

Dunkley et al. (2007),
Larson et al. (2010)

The key evidence

II: The Gunn-Peterson trough

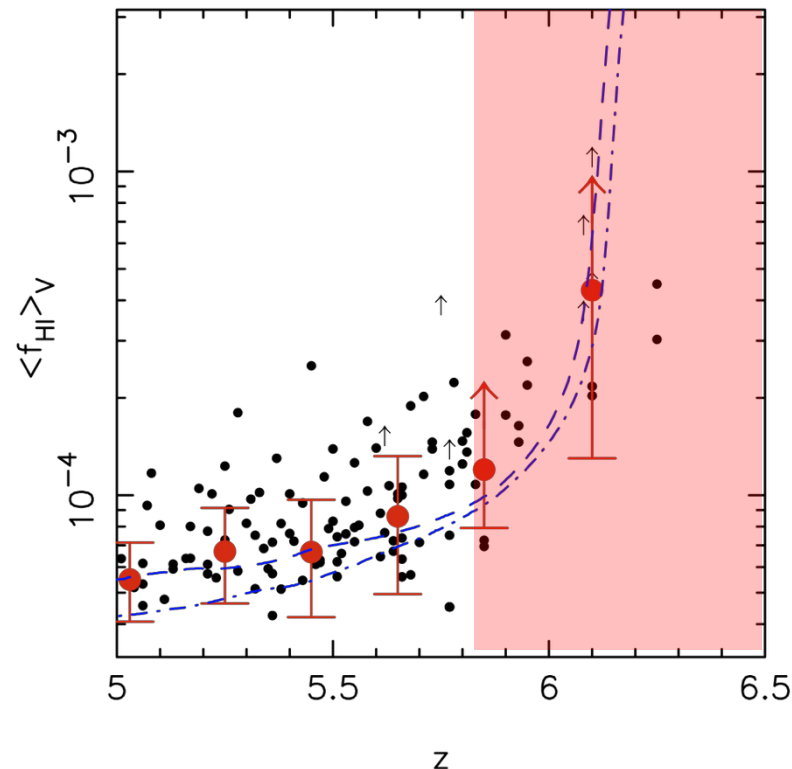


The Ly- α opacity is extremely large even for small neutral hydrogen fractions

Fan et al. (2006)

The neutral hydrogen fraction

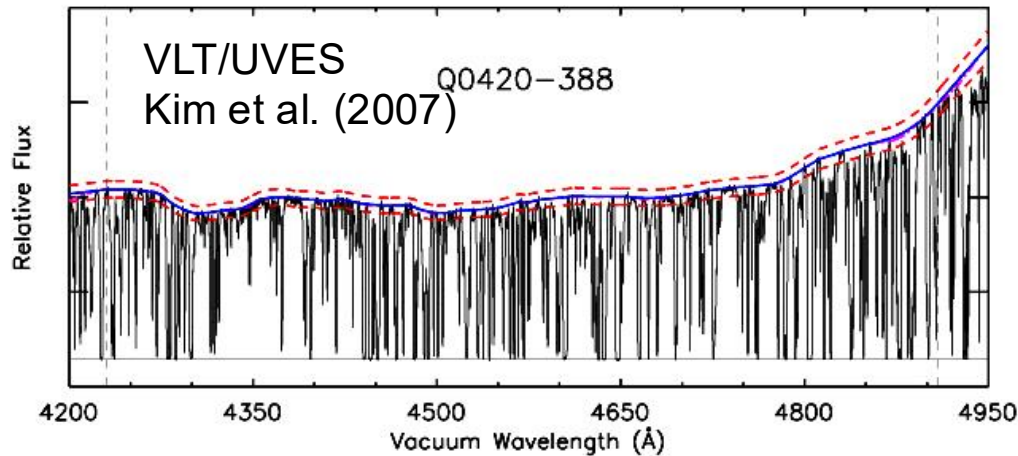
The Gunn-Peterson (1965) trough at $z > 6$ implies f_{HI} is increasing towards higher redshift, but probing the heart of reionisation ($f_{\text{HI}} \sim 1$) directly is impossible.



Fan et al. (2006)

The key evidence

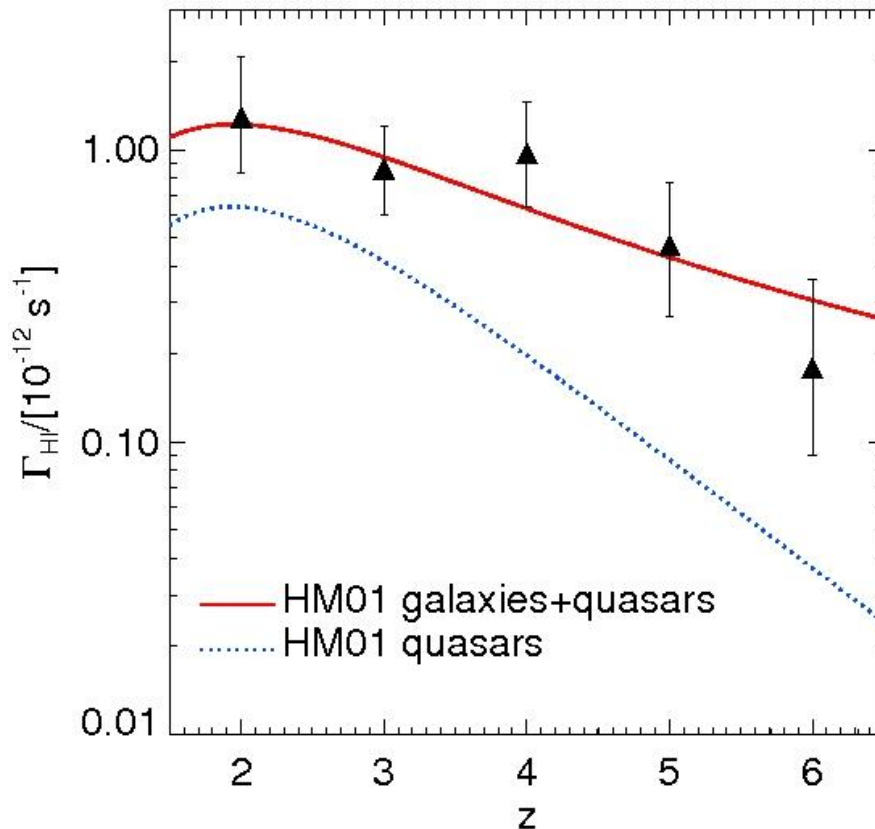
III. Measurements of the H I photoionisation rate at $z < 6$ (the UV background)



Constraints come from measurements of the opacity of the Ly- α forest and the line-of-sight proximity effect.

$$\tau = \tau_0 \frac{(1+z)^6 (\Omega_b h^2)^2}{T_0^{0.7} H(z) \Gamma(z)} (1+\delta)^{2-0.7(\gamma-1)}$$

The contribution of quasars



Bolton & Haehnelt (2007)
see also Faucher-Giguere et al (2008)

- Observed quasars at $z \sim 6$ account for only a fraction of the photons required to reionise the Universe at $z=6$.
- Estimates for the fractional contribution vary from 1-15% (e.g. Srbinovsky & Wyithe 2007).
- An increasingly significant contribution from star forming galaxies required to maintain the UVB photon budget approaching $z=6$.

What we know about reionisation and the UVB:

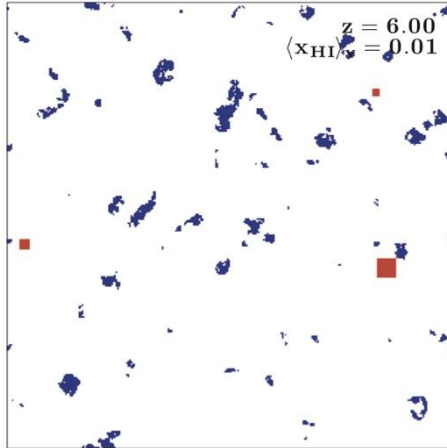
- CMB: integrated reionisation history constraint only. Consistent with $z_r = 10.5$ for instantaneous reionisation. No information on the neutral fraction as a function of redshift, but implies reionisation was extended.
- QALs: appearance of the Gunn-Peterson trough at $z=6$ implies $z_r > 6$, but the interpretation of data with regard to how rapidly the neutral hydrogen evolves around this time is hampered by saturated absorption. Furthermore, it is possible that neutral patches may linger in the IGM even at $z < 6$.
- Quasars are unlikely to reionise the IGM but make an increasingly important contribution to the UVB toward lower redshift. Star forming galaxies appear to provide the bulk of ionising photons at $z \sim 6$ and perhaps earlier too.

Other (controversial) probes of the IGM/EoR using QSO absorption lines

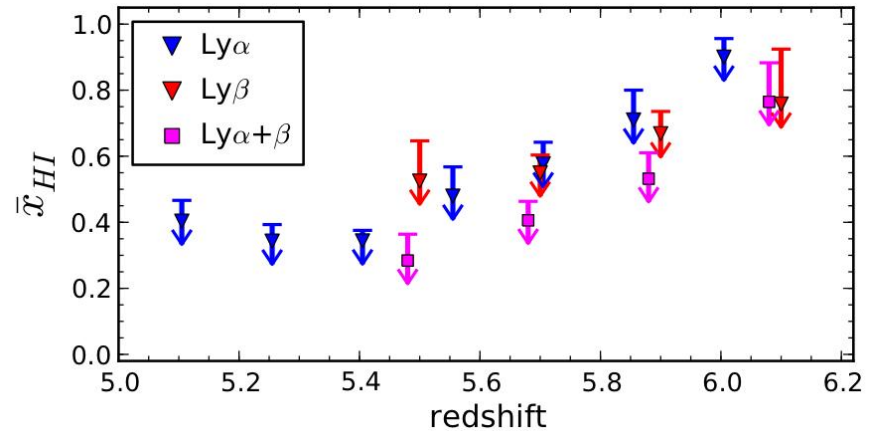
- Quasar near-zones (Wyithe & Loeb 2004, Bolton & Haehnelt 2007, Lidz et al. 2007, Maselli et al 2007, 2009, Alavarez & Abel 2007, Wyithe et al. 2008)
- The GP trough damping wing (Miralda-Escude 1998, Totani et al. 2006, Mesinger & Haiman 2004, 2007)
- Ly- α absorption line widths (Theuns et al. 2002, Bolton et al. 2010)
- Metal absorption lines (Becker et al. 2006, 2009, 2011, Ryan-Weber et al. 2006, 2009, Simcoe 2006, Oppenheimer et al. 2009)
- Dark gap statistics (Songaila & Cowie 2002, Fan et al. 2006, Paschos et al 2005, Gallerani et al 2006, 2008, McGreer et al. 2011)

But: interpretation would be aided by ***larger data sets*** and ***higher resolution***,

Was reionisation even complete by $z=6$?!



Mesinger (2010)



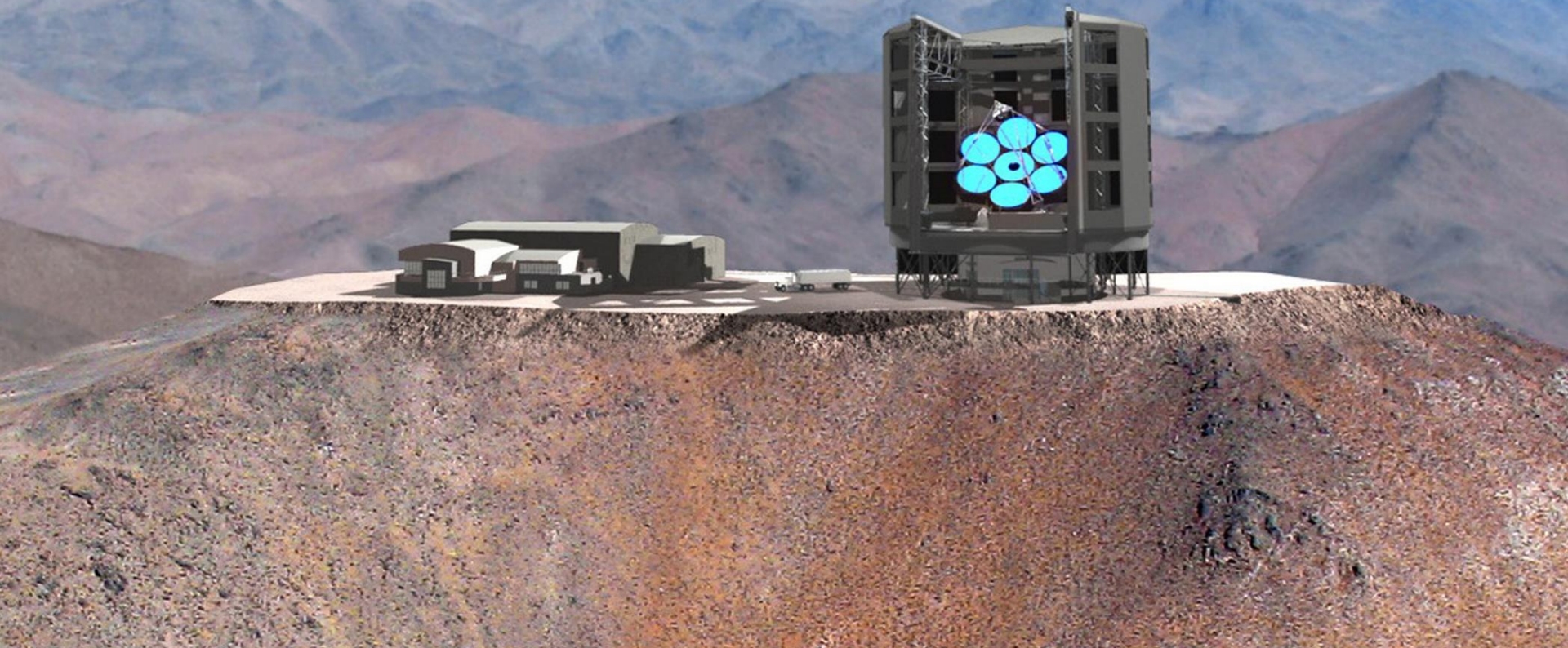
McGreer et al. (2011)

- Sight lines at $z=5-6$ are rare and may simply be too sparse to probe remaining patches of neutral hydrogen.
- Quasars also exist in biased regions and are less likely to pass through neutral regions relative to randomly position sight-lines.
- Based on dark gaps in the Ly- α /Ly- β forest data alone, cannot rule out an IGM which is 10 per cent neutral by volume at $z=5-6$.

Probing reionisation with the GMT

There are two key areas where the GMT will contribute to EoR science:

- 1) Directly observing the first stars/galaxies (S. Finkelstein, K. Freese, A. Pawlik)
- 2) Analysing the impact of these and subsequent sources on the surrounding IGM high resolution optical/NIR spectroscopy.



Advantages of the GMT

The key advantage for optical/near-IR spectroscopy is sensitivity and improved S/N at high spectral resolution.

- Fainter background sources will be accessible.
- Much higher signal-to-noise achievable with high dispersion ($R > 40K$) spectrographs on the brightest sources.

Table 3. First generation instrument candidates.

#	Instrument	λ (μm)	Resolution	FOV	Modes
1	Visible WF MOS	0.4-1.0	500-5000	60-150 sq. arcmin	MOS/IFU/TF/Imager
2	NIR MOS	0.9-2.5	1500-5000	25-100 sq. arcmin	MOS/IFU/TF/Imager
3	Visible Echelle	0.3-1.0	20K-100K	20"	Single Obj/MOS/I cell
4	MIR imager	5.0-24.	5-3000	2' x 2'	AO imager/spectrometer
5	NIR Echelle	0.8-5.0	40K-100K	30"	LTAO/Natural Seeing
6	NIR imager	0.9-5.0	5-5000	30"	ExAO/LTAO imager

G-CLEF

GMTNIRS

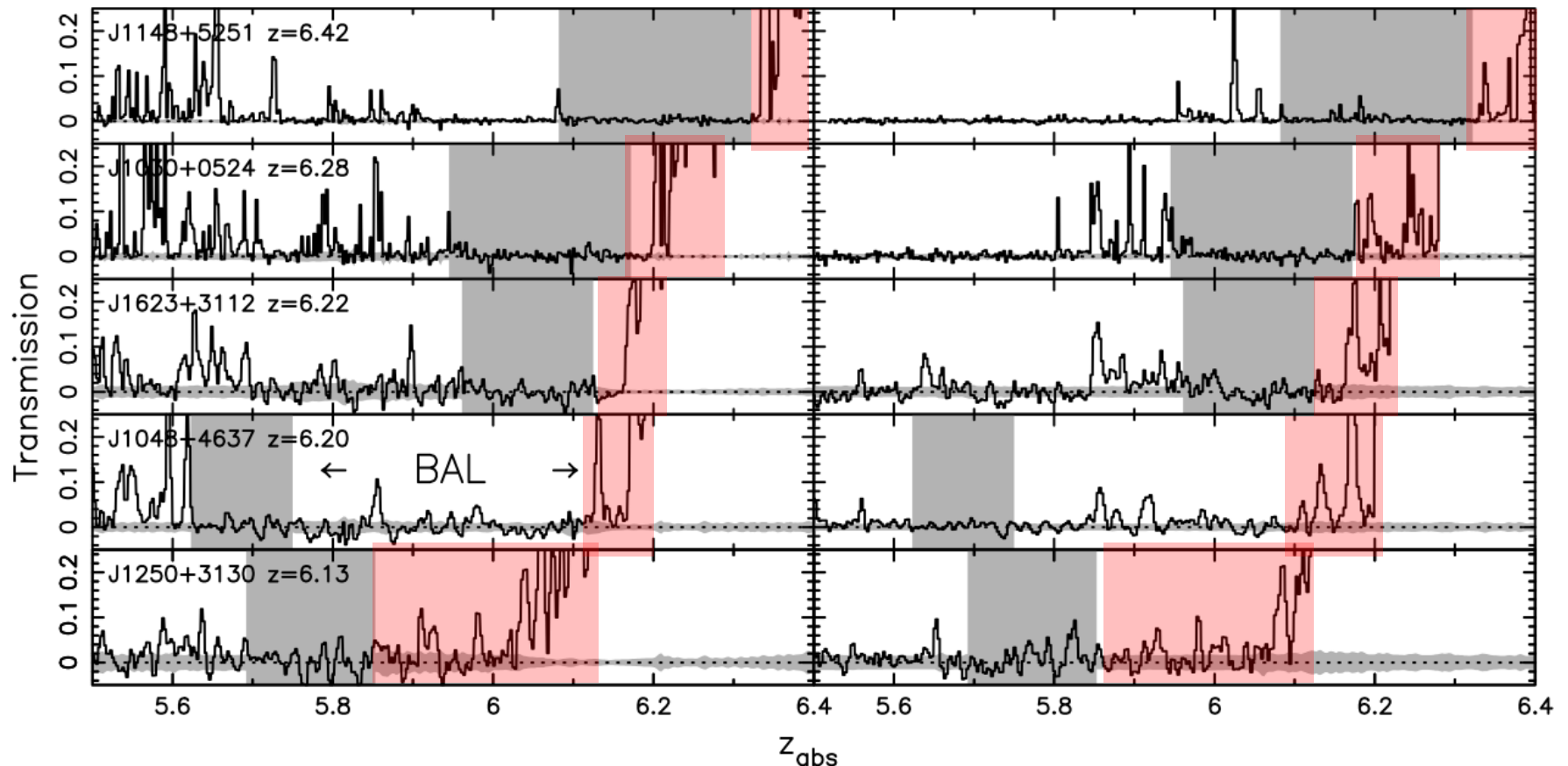
Ref: GMT science technical requirements, <http://www.gmto.org/sciencecase>

Two interesting questions for the GMT

- How does the IGM neutral fraction evolve at $z > 6$?
Quasar near zones as a probe of large neutral fractions in the IGM.
- When did reionisation occur and what kind of sources were responsible?
The thermal state of the IGM as a probe of the reionisation history.

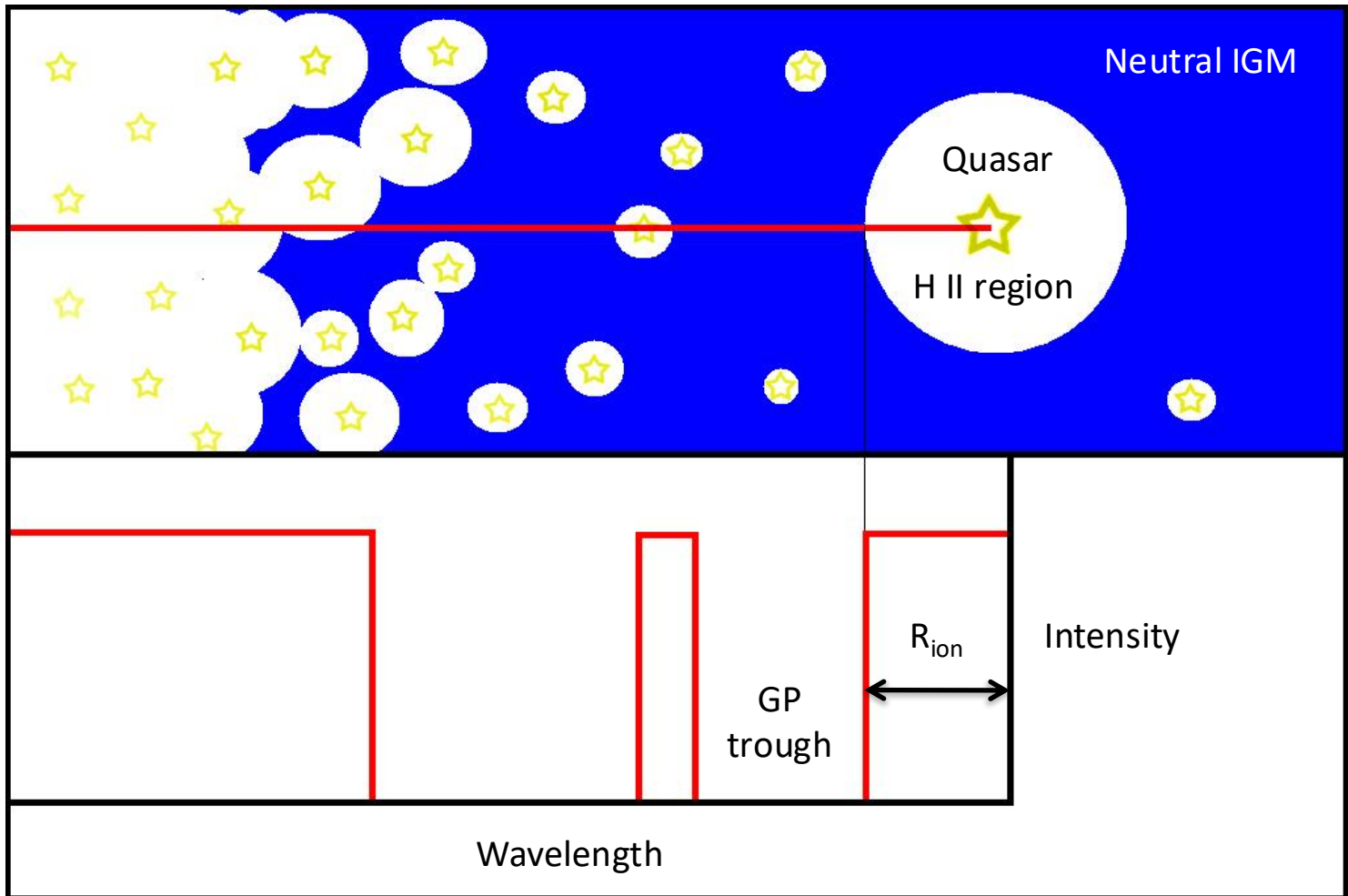
Quasar near-zones

Transmission windows (3-10 proper Mpc) blueward of Ly- α and redward of the Gunn-Peterson trough. Due to enhanced IGM ionisation in close proximity to the QSO.



Fan et al. (2006)

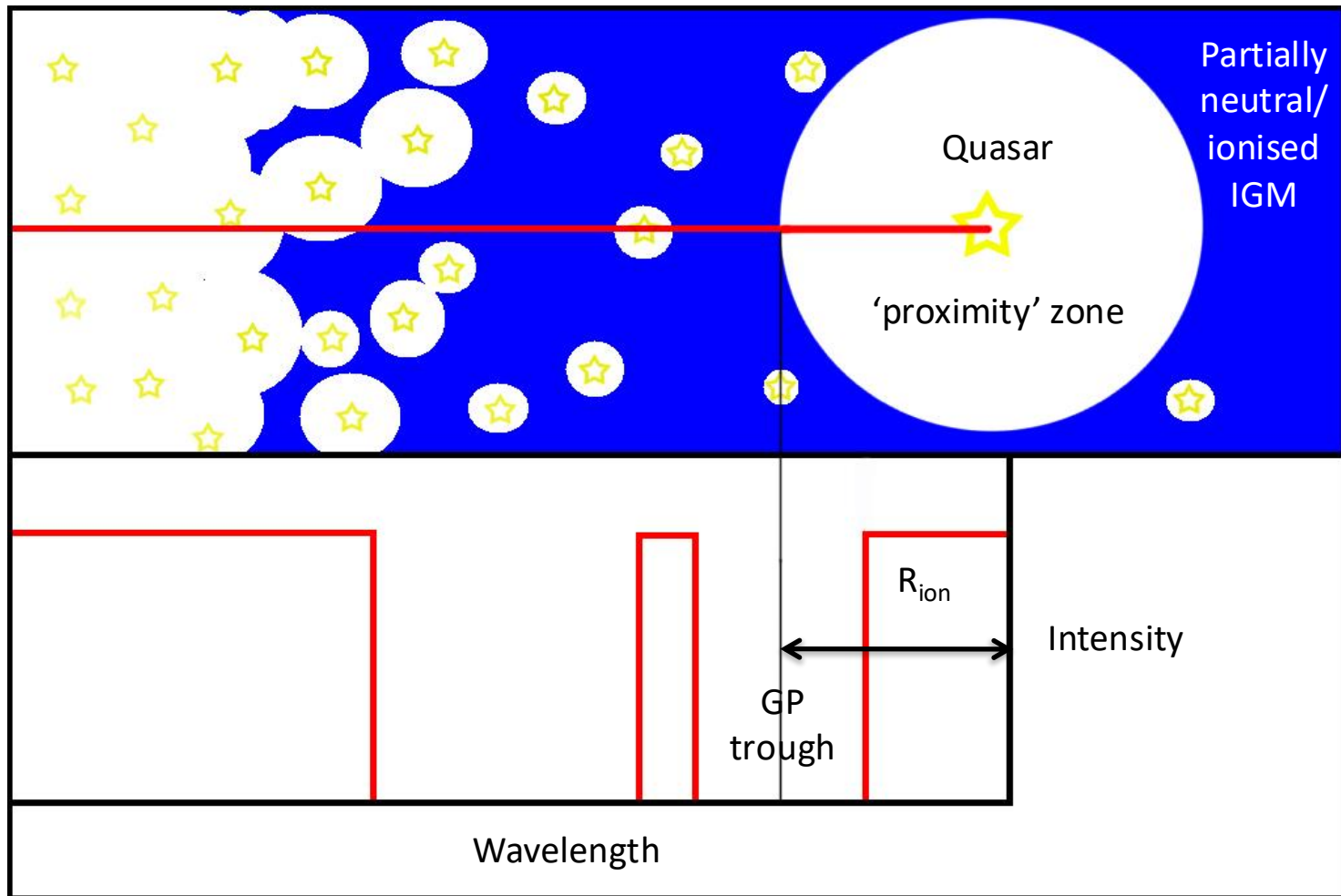
H-II regions...



$$R_{ion} \propto N^{1/3} t_Q^{1/3} f_{HI}^{-1/3}$$

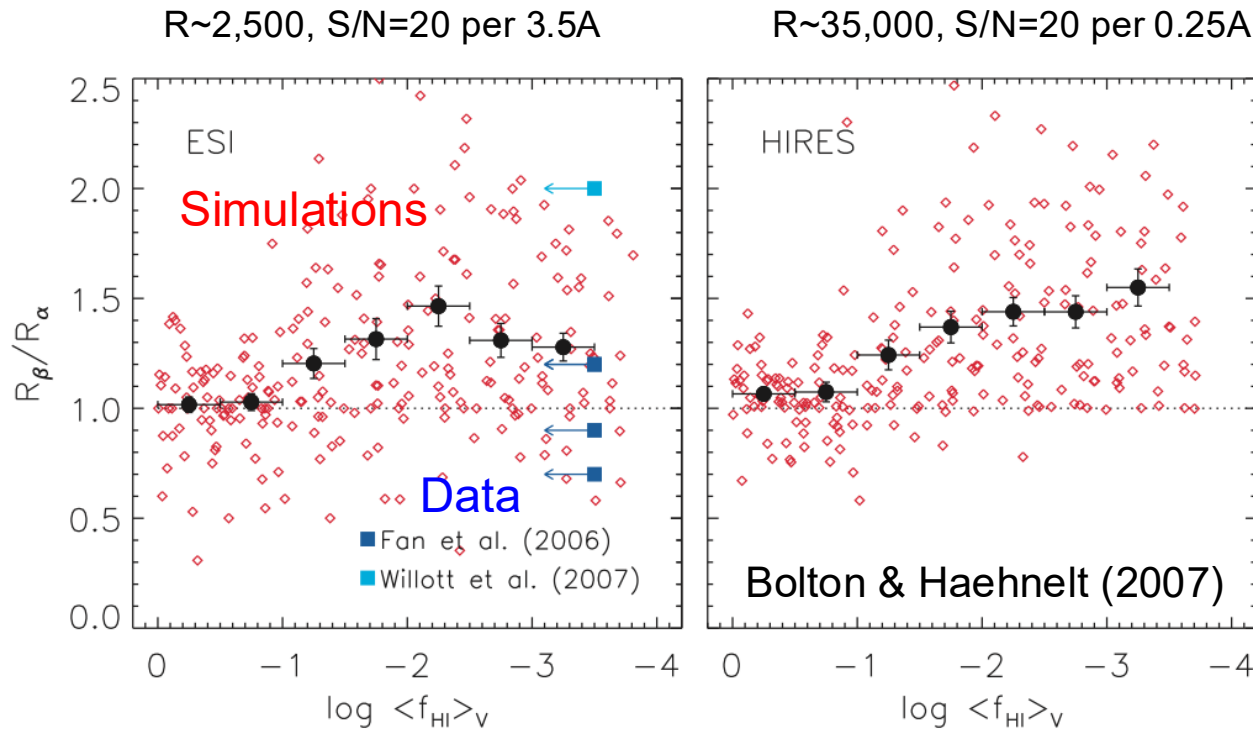
e.g. Wyithe & Loeb (2004)

...or resonant absorption?



$$R_{\alpha} \propto N^{1/2} t_Q^0 f_{HI}^0$$

Near-zone sizes: ratio of Ly- α to Ly- β

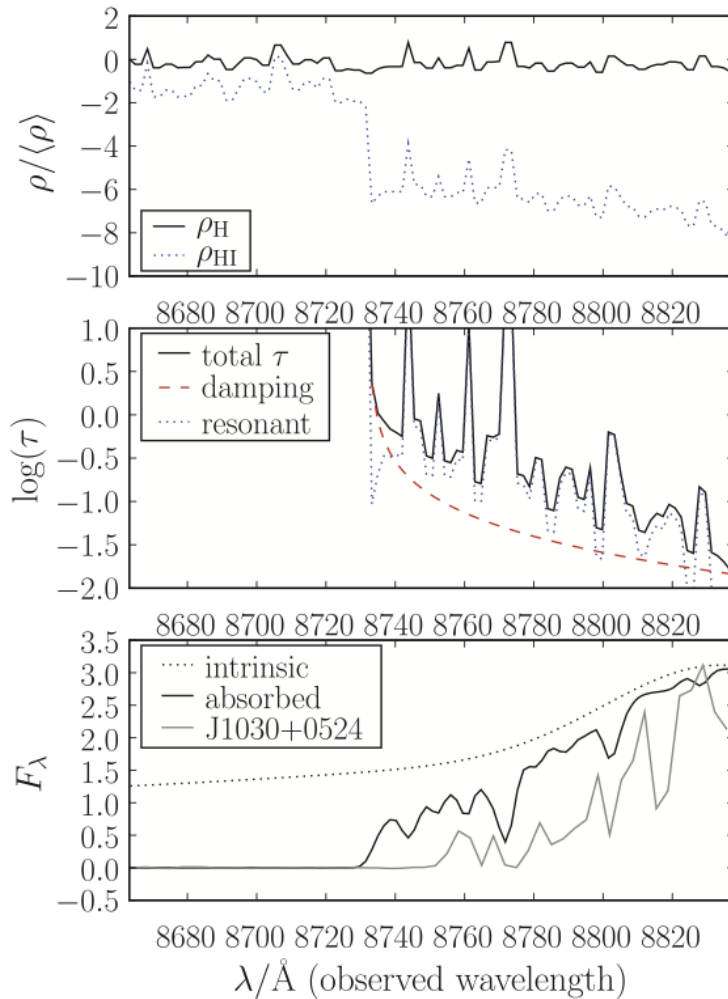


The ratio of the near-zone extent in Ly- β to Ly- α is sensitive to $f_{\text{HI}} > 0.1$

But: current data at $z > 6.1$ do not provide any constraining power.

More spectra (> 30 for 3σ) needed and **higher resolution** is desirable.

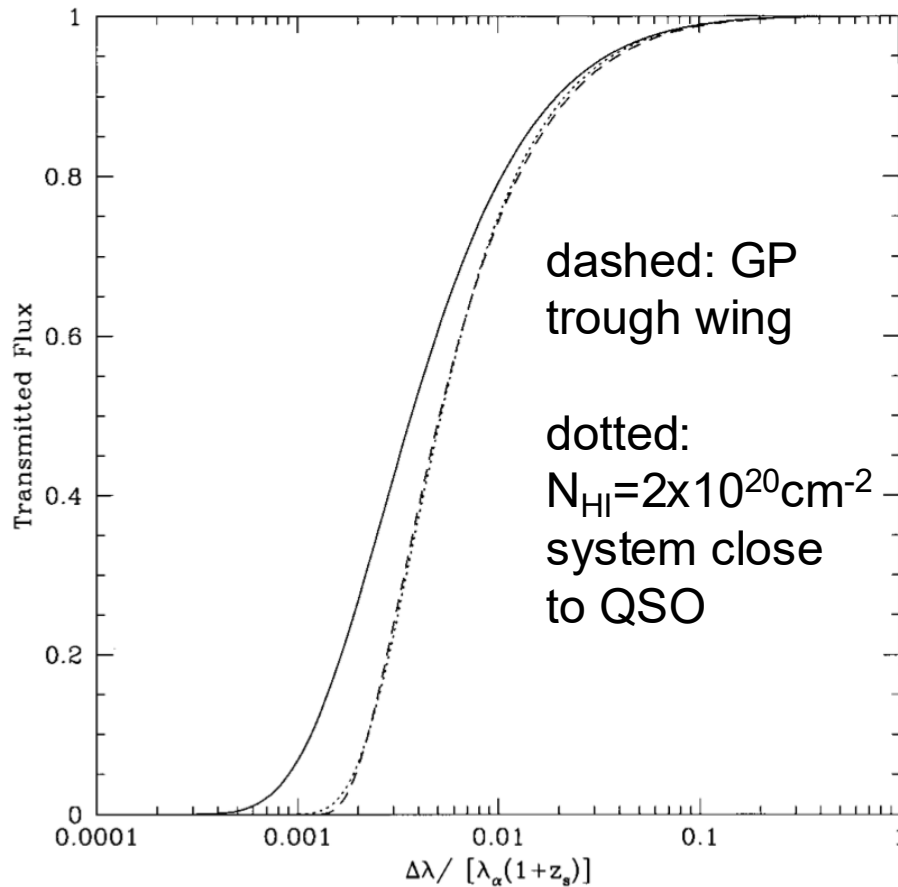
The GP trough damping wing



- Significantly neutral IGM.
- Transmission redward of the GP trough edge expected to be smoothly attenuated by a damping wing.

Kramer & Haiman (2009)

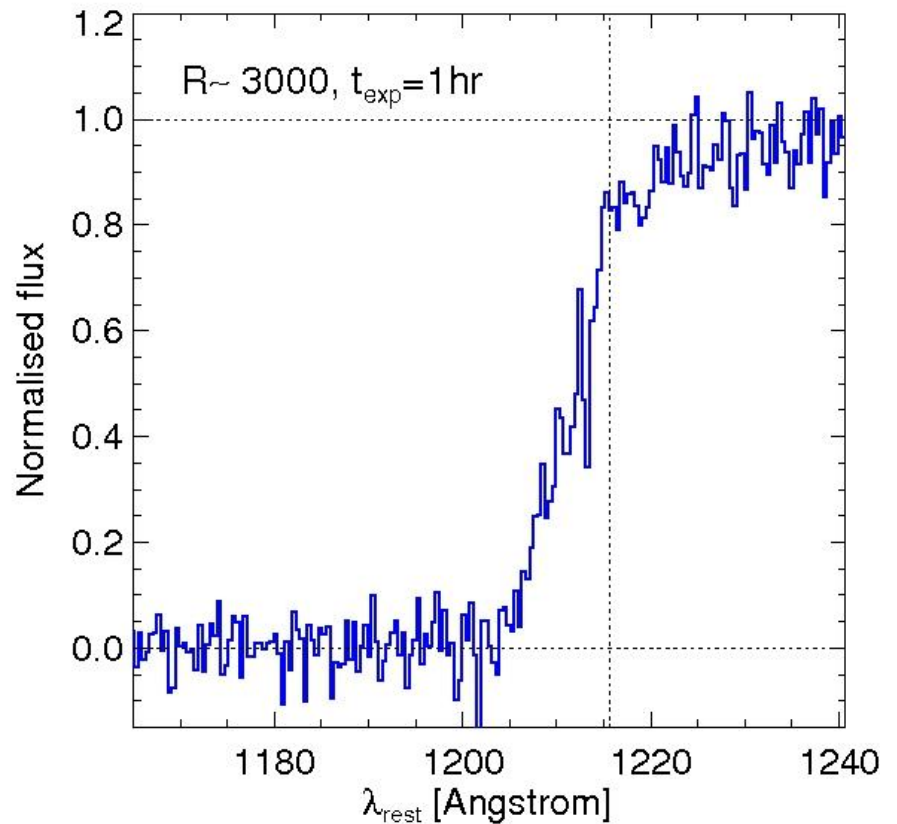
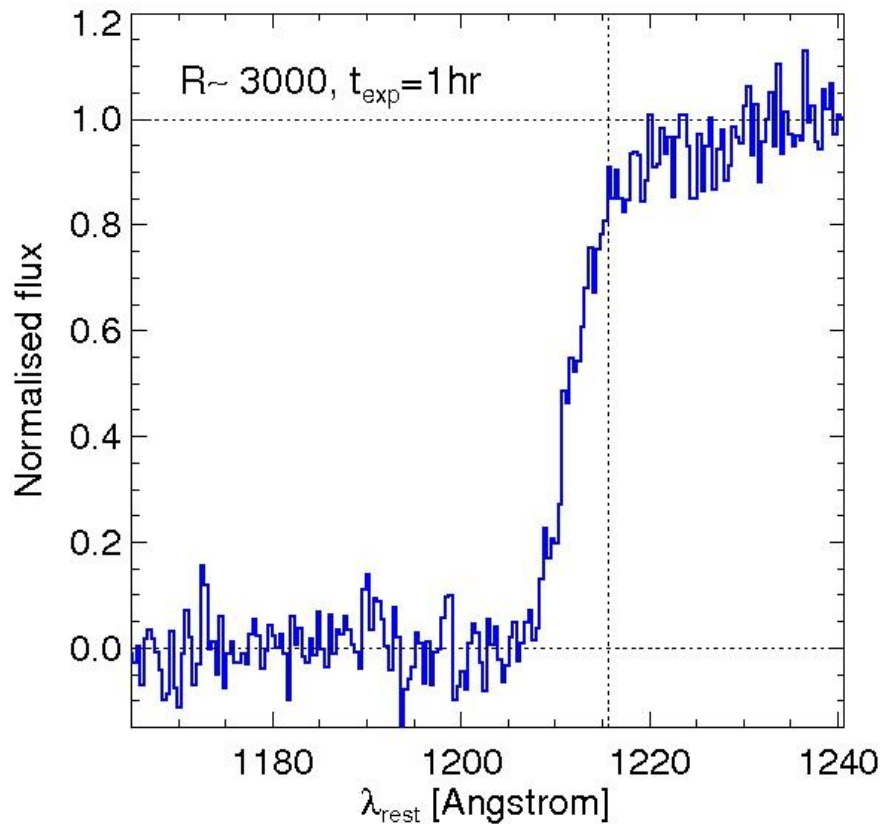
The GP trough damping wing



But: difficult to distinguish between a GP damping wing vs. high column density absorption system.

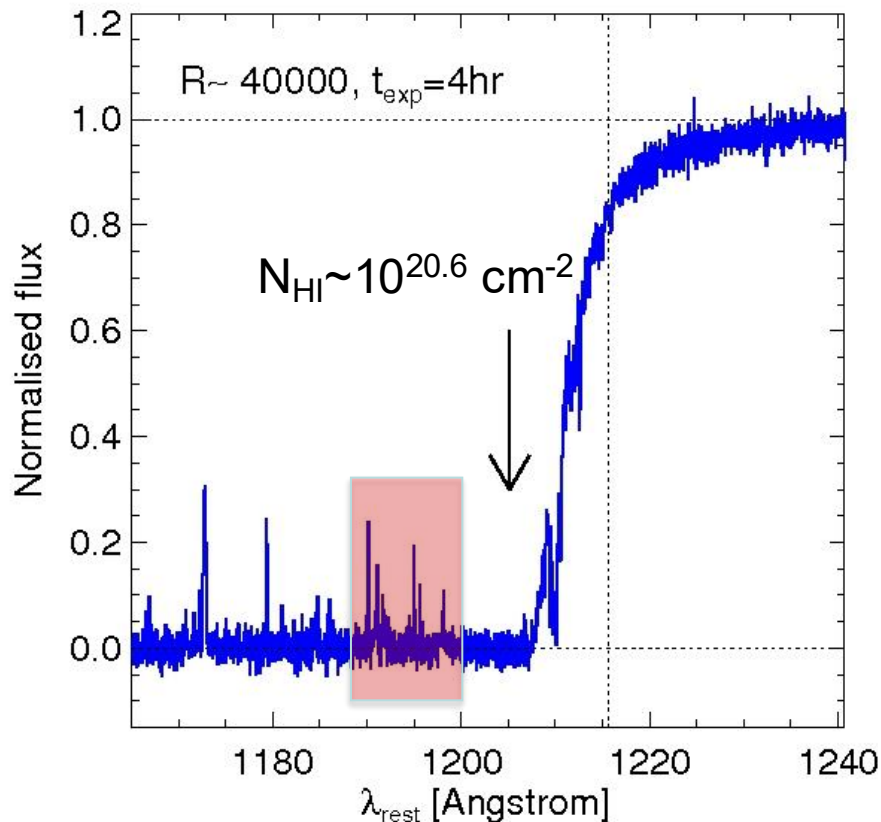
Miralda-Escude (1998)

Simulation- Keck/ESI

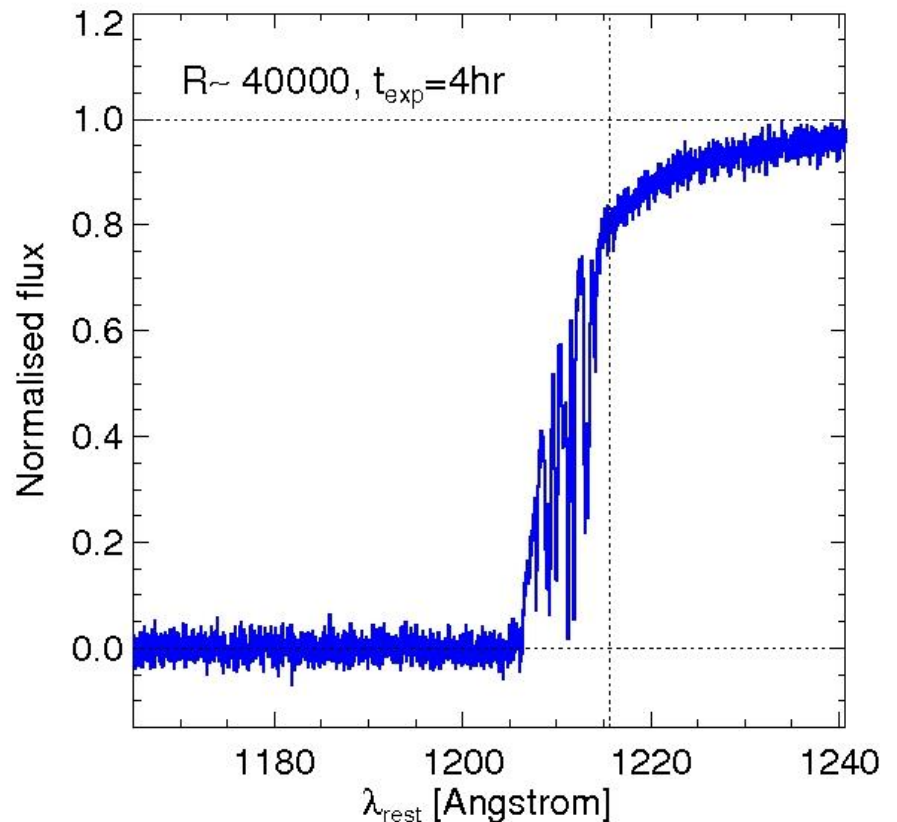


$i \sim 22$, $S/N = 20$ per pixel (e.g. Fan et al. 2006)

Simulation – GMT/G-CLEF



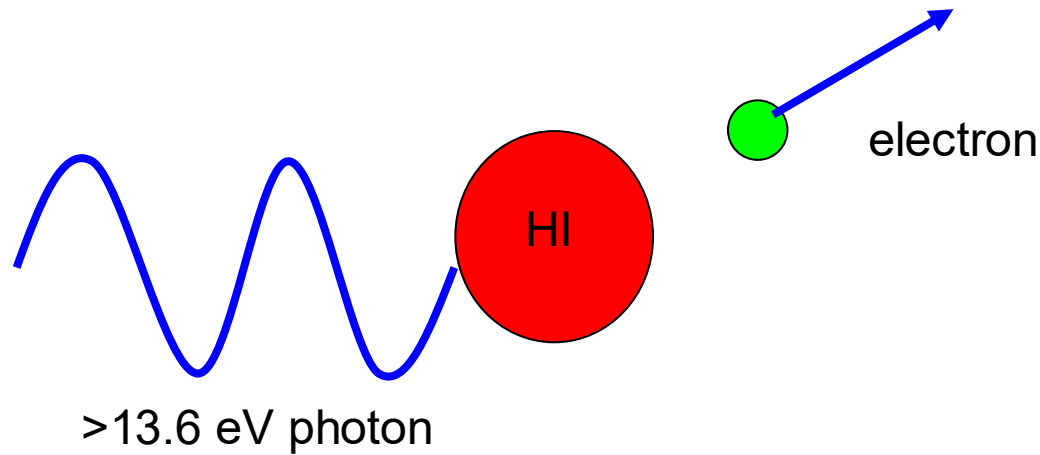
DLA lying close to the QSO
IGM highly ionised



GP trough damping wing
IGM neutral

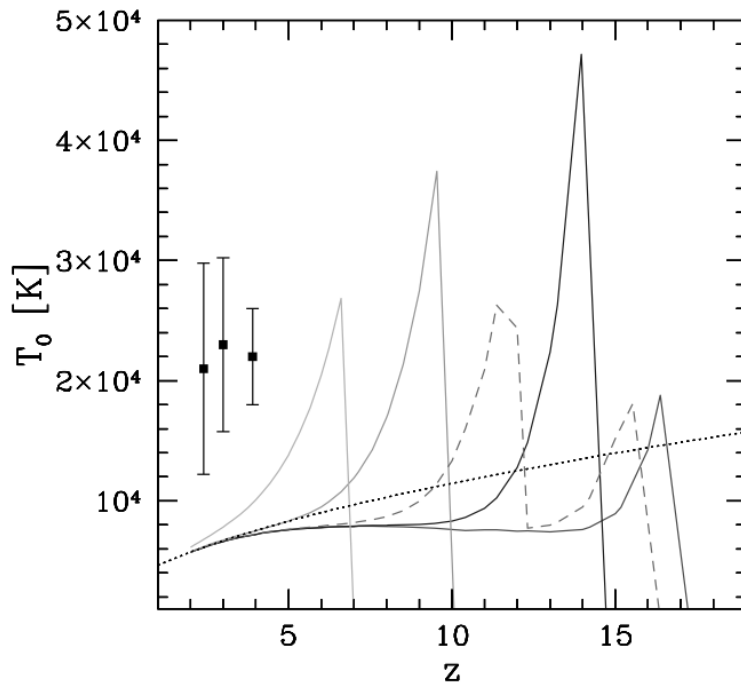
Photo-heating

Photons not only ionise – if they have $E > E_{\text{th}}$ ($\text{H I} = 13.6\text{eV}$) then they also heat the IGM.



Electrons share their energy with the baryons via Coulomb scattering and raise the temperature.

Reionisation and the thermal history



Hui & Haiman (2003)
see also Theuns et al. (2002)

- The temperature of the low density IGM provides an indirect probe of the reionisation history.
- Long cooling timescale enables use as an indirect probe of the H I and He II reionisation epochs. The IGM retains a 'thermal memory' of reionisation (until the thermal asymptote).

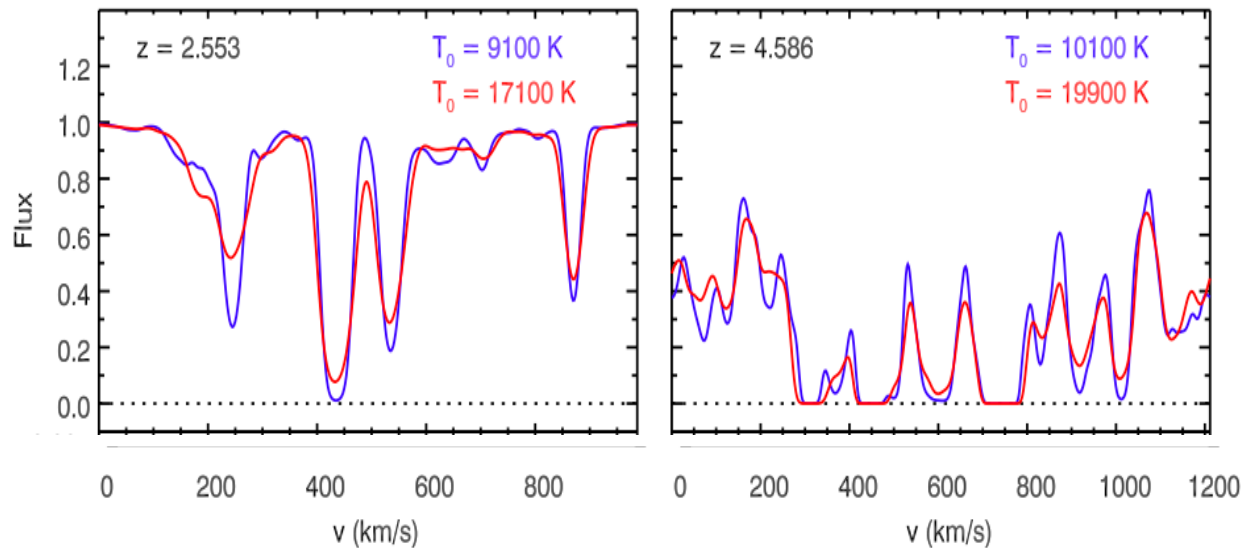
The temperature of the IGM thus depends on:

- 1) When the IGM was reionised (how much time available to cool?)
- 2) The spectra of the ionising sources responsible for reionisation (harder spectra = more heating).

The IGM temperature

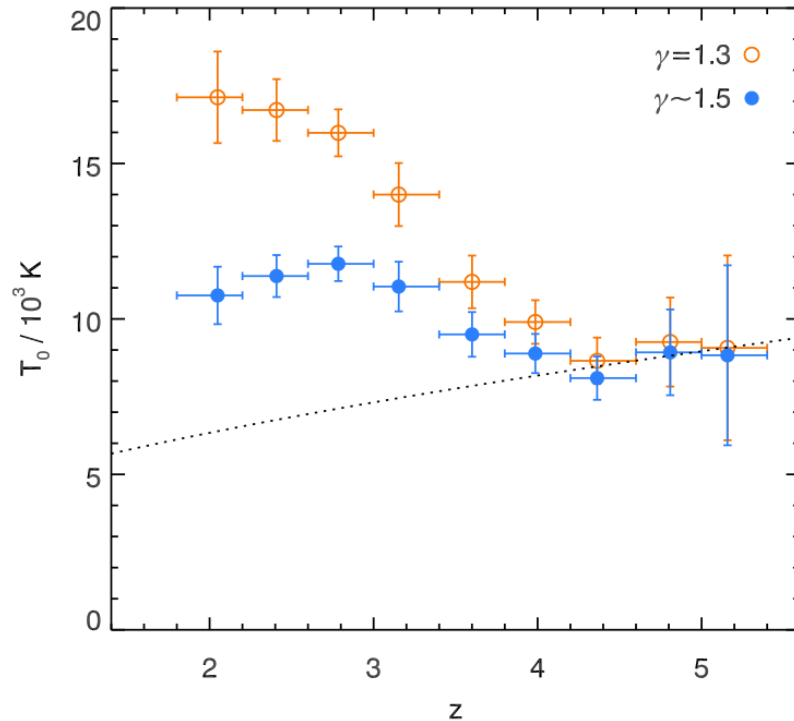
Higher temperatures broaden absorption features through:

- 1) Thermal broadening due to the *instantaneous temperature* (along the line of sight.)
- 2) Jeans (pressure) smoothing due to the *integrated heating history* (in three dimensions.)



Becker et al. (2011)

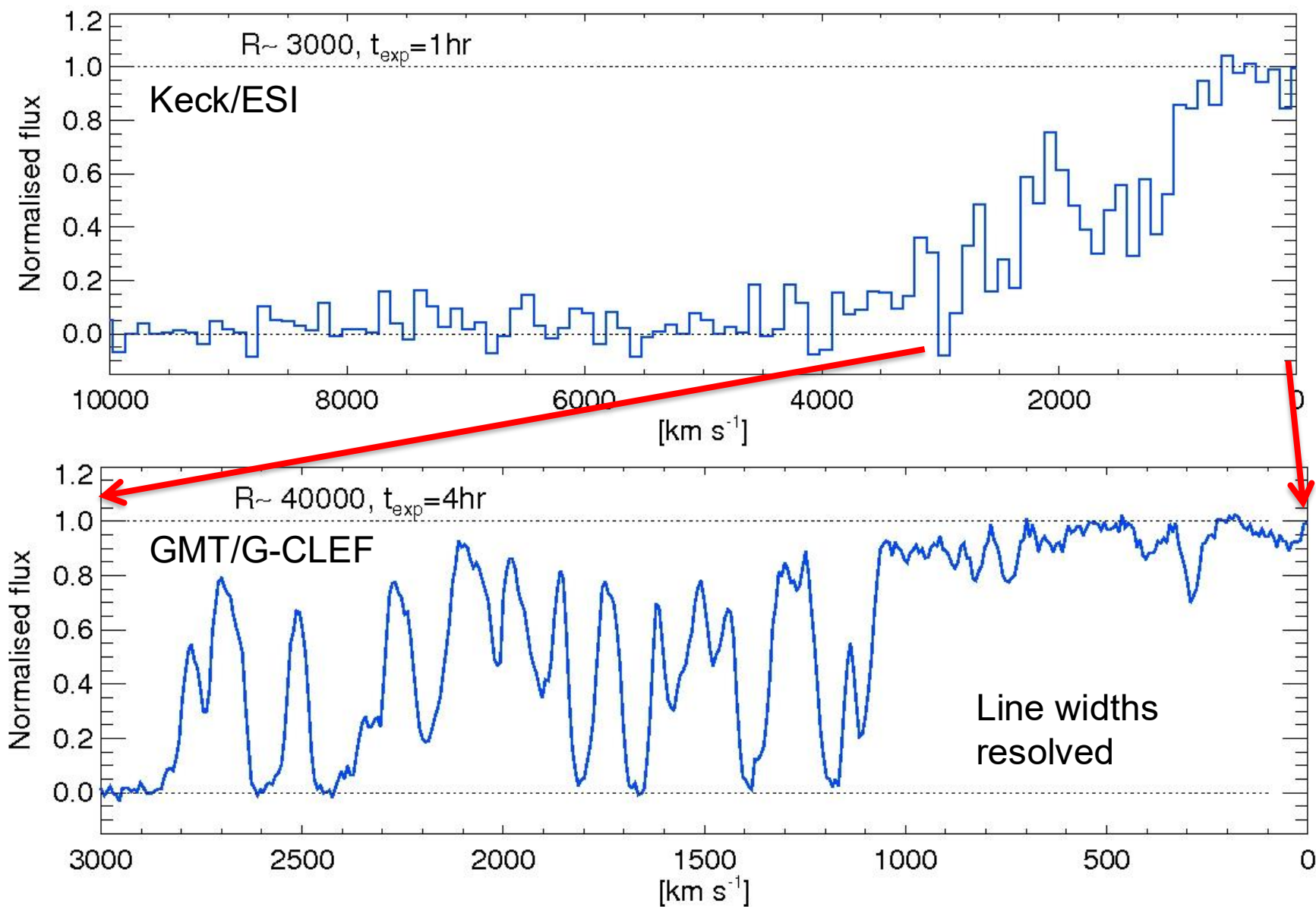
IGM temperature at $z < 5$



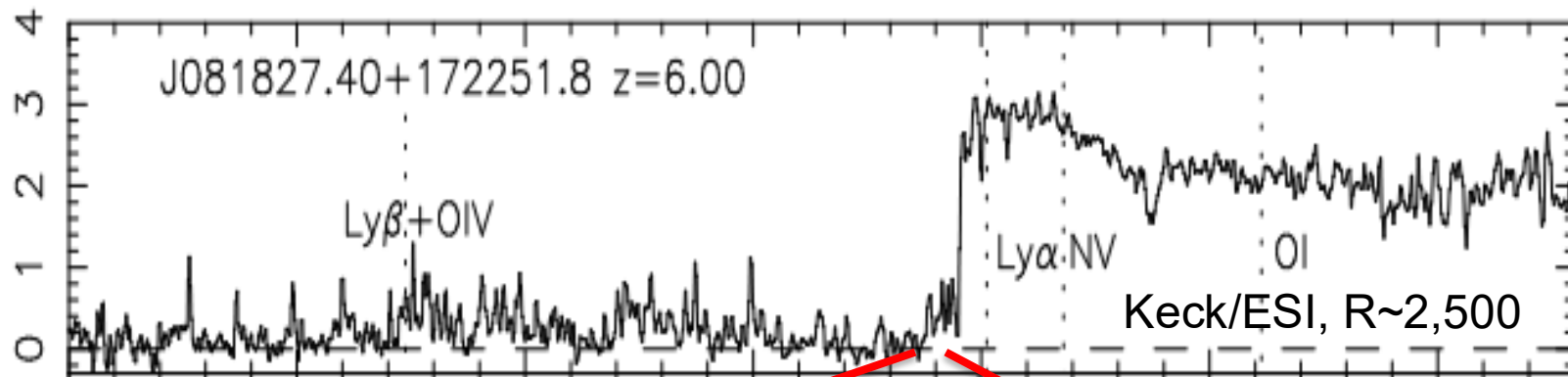
Becker et al. (2011)

- A clear rise in temperature towards $z=2$, unambiguous signature of He-II reionisation at $z < 4.5$.
- However, He-II heating complicates interpretation with respect to H I reionisation.
- Higher redshift needed to push beyond He II reionisation and the thermal asymptote for H I reionisation.

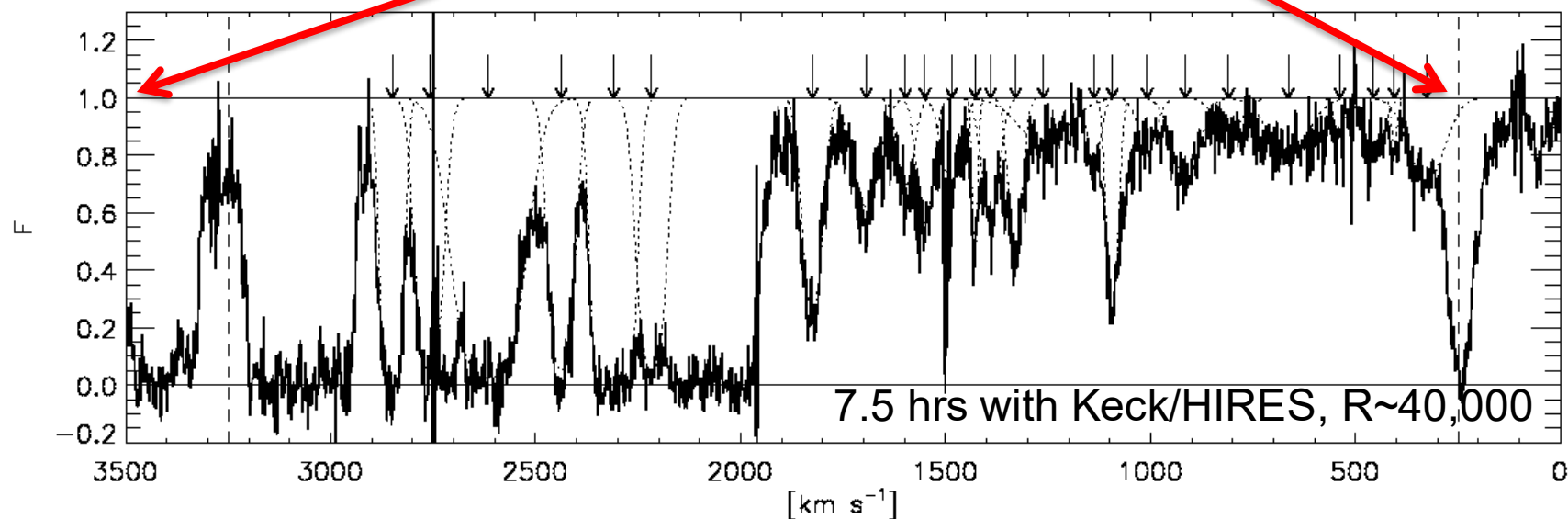
IGM temperature at $z > 6$



IGM temperature at $z=6$



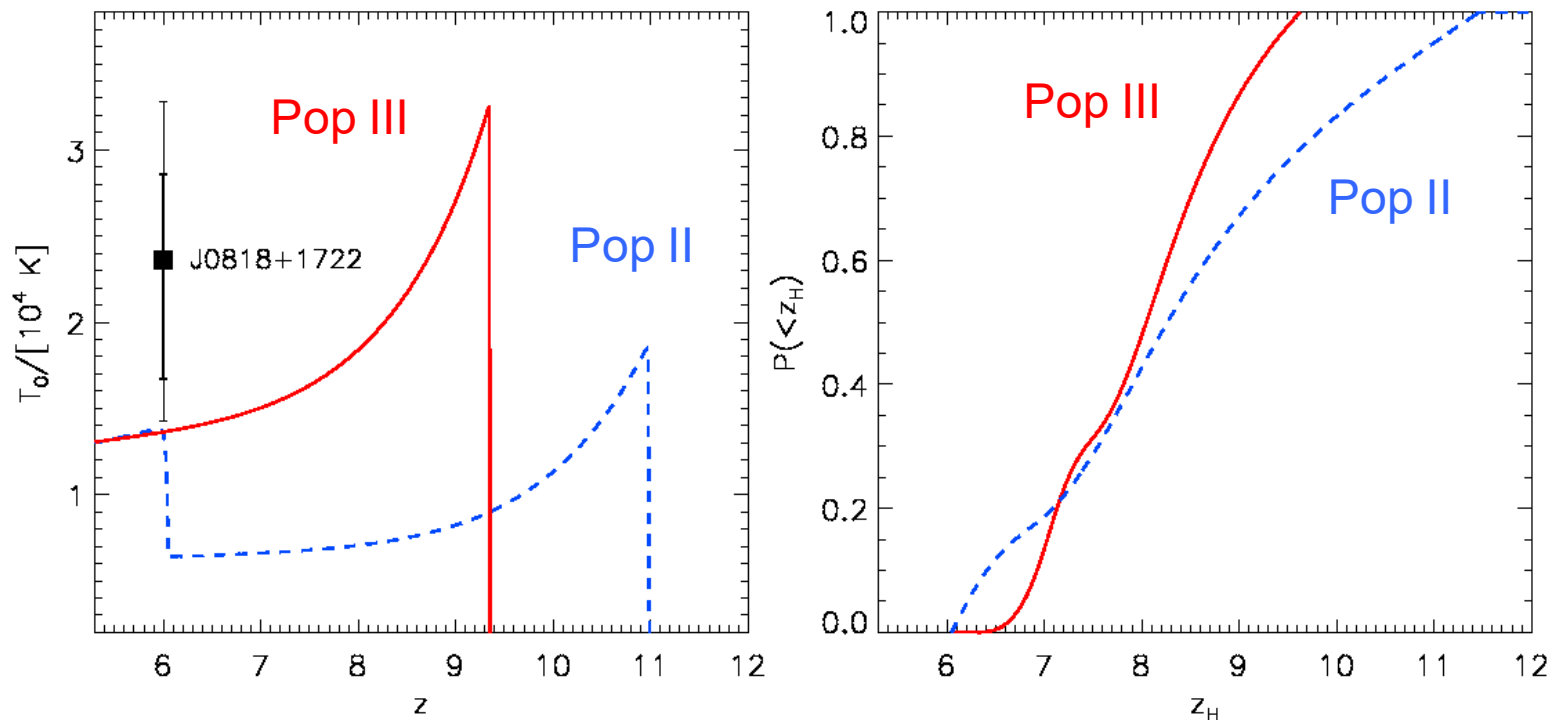
Fan et al. (2006)



Becker et al. (2007), Bolton et al. (2010)

A thermal constraint on reionisation

- Heating depends on intrinsic spectra of the first sources
- Current measurement from a single HIRES spectrum at $z=6$ is dominated by statistical uncertainty.



Bolton et al. (2010)

Summary

- Optical/NIR echelle spectroscopy with the GMT can provide **high resolution, high signal to noise data** (QSOs/GRBs) which will probe the physical state of the IGM at $z > 6$ with unprecedented detail.
- Spectra from fainter background sources also enables **more sight-lines and improved statistics** for somewhat lower signal-to-noise.
- The **sizes of quasar “near-zones”**, detection of the **GP trough damping wing** will enable insight into the epoch when the IGM was largely neutral.
- Measurements of the **IGM temperature at $z > 6$** are sensitive to the spectra of the ionising sources and the timing of reionisation.
- Plus much more (metals, IGM tomography, dark gap statistics...)